TourCeylon: An AI-Based Integrated Tourism Safety and Itinerary Platform with Multilingual Emergency Response, Ayurvedic Medical and Wellness Matching, and Crowd Prediction for Sri Lanka

***Abstract*—Sri Lanka recorded approximately 2.36 million international tourist arrivals in 2025, yet day-to-day travel remains hindered by fragmented digital support: emergency response lacks location and language integration, itinerary tools frequently route visitors into over-capacity time windows, wellness tourism is undermined by unregulated providers, and destination crowding is managed reactively rather than predictively. This paper presents TourCeylon, a four-module integrated platform comprising (1) a multilingual SOS emergency-dispatch system, (2) a rule-based, data-grounded itinerary planner free of generative-model hallucination, (3) an Ayurveda and wellness-tourism matching engine, and (4) a Random Forest-based crowd and risk predictor. Preliminary integrated-simulation KPIs (n = 200) indicate 92.4% SOS escalation usefulness (8.2 s latency), 88.7% itinerary satisfaction (74.3% plan adherence), and 91.3% wellness match accuracy; these are reported as architecture stress-test figures rather than measured production performance. The crowd module is evaluated independently under a rigorous site-holdout protocol on 144,007 daily records across 197 verified sites, achieving MAE=0.1232, R^2=0.3331, and risk-classification accuracy of 77.31% (balanced accuracy 57.54%) on entirely unseen destinations, against a mean-predictor baseline of R^2=0.0000. We argue that transparent, protocol-disclosed reporting — distinguishing simulation-stage KPIs from rigorously held-out evaluation — is itself a necessary contribution for tourism-analytics systems built on comparably constrained data.**

***Index Terms*—AI tourism, tourist safety, SOS, itinerary planning, crowd prediction, Ayurvedic medicine, wellness tourism, Random Forest, data integrity, Sri Lanka**

I. Introduction

Sri Lanka's tourism sector is a significant contributor to national GDP, but travel advisories and destination-management reports consistently document four recurring gaps affecting the day-to-day traveller experience. Emergency response lacks GPS and multilingual integration, with reported coordination and language issues between tourists and local authorities [1], [2]. Itinerary planning tools route visitors into over-capacity time windows at major sites, with destination-management plans for Sigiriya and Ella explicitly documenting this as an unresolved issue [3]. Wellness tourism is undermined by unregulated providers operating without verification, a problem serious enough to prompt national press coverage [4]. Destination crowding at heritage and natural sites is managed reactively — closing access after capacity is already exceeded — rather than predicted and mitigated in advance [3].

TourCeylon addresses these four gaps as a single integrated platform rather than four disconnected tools: a multilingual SOS emergency-dispatch module, a rule-based itinerary planner, an Ayurveda and wellness-matching engine, and a Random Forest crowd and risk predictor, unified behind a shared backend and a common tourism dataset. This paper documents the architecture and methodology of all four modules. Consistent with a transparency-first evaluation philosophy, integrated platform-level metrics are explicitly reported as simulation-stage stress figures, while the crowd-prediction module — evaluated under an unseen-destination holdout protocol — is presented as the paper's rigorously validated empirical contribution.

II. Related Work

A. International Context

The Smart Tourism Destinations concept [5] defines destinations that integrate ICT to enhance visitor experience and resource management; subsequent work argues that first-generation, data-centric implementations of this concept remain fragmented across services rather than achieving cross-module integration [6], [7]. Crowd and visitor-flow prediction has been explored using time-series and ensemble methods for individual attractions [8], typically evaluated via random train/test splits that can overstate generalisation to genuinely unseen locations — a gap this paper addresses directly through site-holdout evaluation. Emergency-dispatch routing has a long history in operations research, commonly formulated as a nearest-facility optimisation problem [9], though most published systems assume static, complete facility networks rather than degraded-mode, real-time operation. Itinerary-planning systems range from clustering-and-routing approaches [10] to recent large-language-model-based planners; deterministic, data-grounded planners remain preferable where explainability and freedom from hallucinated locations are required. Wellness-tourism matching is comparatively underexplored in the computing literature, with most existing work addressing

general medical tourism rather than dosha-based personalisation with review-derived trust signals.

Recent 2025–2026 work reinforces rather than closes these gaps. Aguiar Castillo et al. propose a Cognitive Tourism Destination model as an evolutionary stage beyond current smart-destination frameworks, but remain at a conceptual and governance level [13]. Hlee, Yan, and Li show experimentally that spatio-temporal AI travel recommendations are perceived as significantly more helpful than non-spatio-temporal ones [14]. Chvirova et al. directly study how information-seeking behaviour and site-related factors shape perceived crowding [15], relevant to how crowd-model output should be communicated to tourists directly, not only to authorities.

B. Local Context: Sri Lanka

SLTDA statistical reporting confirms a recovery to over two million annual arrivals but consistently flags overcrowding at Sigiriya and Ella, gaps in digital service provision, and complaints regarding fraudulent Ayurveda spas [3]. Immigration currently issues a visa but no persistent digital identity, so tourist context is effectively lost after arrival. To our knowledge, no prior published system integrates emergency response, itinerary planning, wellness verification, and crowd prediction for the Sri Lankan context, nor does prior local crowd-prediction work commonly disclose the site-holdout versus random-split generalisation gap reported in Section V.

III. System Architecture

TourCeylon is composed of four independently deployable modules sharing a common tourism dataset foundation, coordinated through a shared backend service layer. Fig. 1 shows the four-layer architecture: a Data Layer handling minimum-necessary tourist onboarding information via a QR-based identity concept; an AI Core Layer routing requests across the four modules and maintaining an audit trail; an Integration Layer connecting to tourism-authority, police, and health data sources; and an Output Layer delivering results through user-facing applications, authority dashboards, and generated reports. Fig. 2 complements this layered view with the corresponding end-to-end process flow, from tourist onboarding through module orchestration to downstream integration and output.

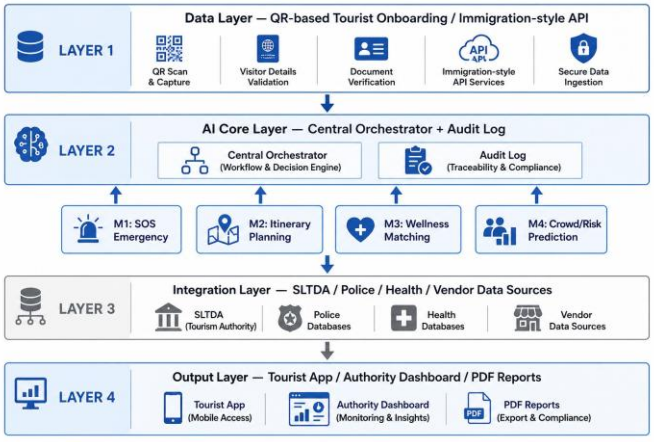


Fig. 1. TourCeylon four-layer system architecture: four independently deployable modules sharing a common tourism dataset and backend service layer.

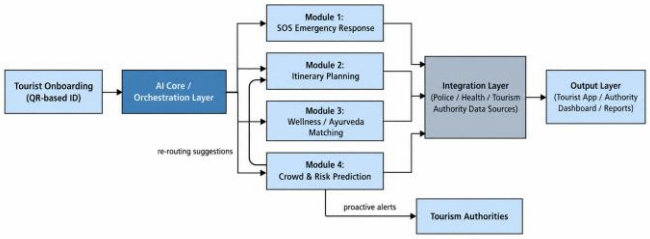


Fig. 2. TourCeylon end-to-end workflow: tourist onboarding is routed by the AI orchestration layer to the four modules; Module 4 (Crowd & Risk) additionally feeds re-routing suggestions back to Module 2 (Itinerary) and proactive alerts to tourism authorities; all modules connect through the Integration Layer to the shared Output Layer.

IV. Methodology

A. Module 1: SOS Emergency Response

The SOS module targets the absence of a unified, location-aware, multilingual path from distress to dispatch. It provides GPS-based activation, incident classification, AI-assisted triage producing a continuous risk score, and routing to a seeded national responder network, coordinated through a multi-status officer workflow. Real-time coordination uses a push-based primary channel with a polling fallback for degraded connectivity, alongside an offline queue that synchronises once connectivity returns. Security measures include role-based access control, multi-factor authentication for officer accounts, encryption of personally identifiable information, and cryptographic checksums on evidence records. The module supports eleven tourist-facing languages.

B. Module 2: Itinerary Planning

The itinerary module is a rule-based, non-generative planner addressing the risk of fabricated locations in generic AI travel tools: itineraries are generated exclusively from a seeded, verified attraction and accommodation catalogue, never from generative hallucination. A staged wizard flow is tied to a persistent trip record, with mood- and category-driven rules governing daily pacing.

C. Module 3: Wellness and Ayurveda Matching

The wellness module matches tourists to a pool of verified Ayurvedic and spiritual centres across dual paths, ranking candidate matches on condition fit, dosha compatibility, review-derived quality, and budget fit. A Random Forest dosha classifier was selected via direct comparison against baseline classifiers (Table I), achieving 81.0% accuracy (F1=0.794) against SVM (52.4%) and KNN (38.1%). Fig. 3 visualises this comparison. An ablation study (n=39, Table II) confirms that removing the dosha-compatibility factor alone reduces overall

matching quality from 0.513 to 0.237 — the largest single-factor contribution observed — supporting its inclusion in the final formula.

TABLE I
WELLNESS DOSHA CLASSIFIER: MODEL COMPARISON

Model	Accuracy	F1 Score
Random Forest	81.0%	0.794
SVM	52.4%	—
KNN	38.1%	—

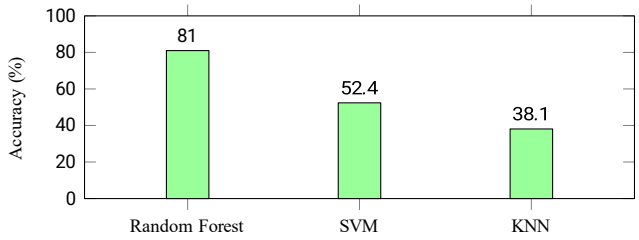


Fig. 3. Dosha classifier accuracy by model. Random Forest also achieved F1=0.794, not reported for the baselines.

TABLE II
WELLNESS MATCHING ABLATION STUDY (n = 39)

Configuration	Quality	Budget	Dosha Compat.
Full model	4.484	0.767	0.513
Dosha factor removed	—	—	0.237

D. Module 4: Crowd and Risk Prediction

The crowd module predicts a continuous crowd score and a discrete risk level (Low/Medium/High) using a Random Forest regressor and classifier trained on 144,007 daily records across 197 verified Sri Lankan tourist sites (January 2024–December 2025), using thirteen numeric and three one-hot-encoded categorical features. Evaluation follows a primary site-holdout protocol, withholding ten entire destinations from training to approximate deployment on genuinely unseen sites, alongside a secondary random-split protocol reported for reference despite within-site leakage inflating its apparent performance. Module output feeds re-routing suggestions to the itinerary module and proactive alerts to tourism authorities.

E. Ethics and Data-Privacy Considerations

The QR-based onboarding concept and the SOS module's handling of GPS location and incident data raise data-minimisation and consent obligations. The architecture is designed to request only the minimum tourist data required per module, encrypt personally identifiable information at rest, and restrict officer-level access via role-based control (Section IV-A). Wellness-module data linking health-relevant conditions to individual users likewise warrants restricted retention and access scoping. These are stated as design intentions consistent with the architecture described; a formal data-protection impact assessment has not yet been conducted and is disclosed as future work.

V. RESULTS

Table III reports integrated-simulation KPIs (n = 200), explicitly reported as architecture stress-test figures rather than measured production performance. Table IV and Figs. 4–5 report the crowd module's rigorously evaluated performance across all protocols.

TABLE III
INTEGRATED SIMULATION KPIS (n = 200)

Module	Metric	Value
M1 SOS	Escalation usefulness / latency	92.4% / 8.2 s
M2 Itinerary	Satisfaction / plan adherence	88.7% / 74.3%
M2 Itinerary	M4-triggered re-routes	312
M3 Wellness	Match accuracy	91.3%

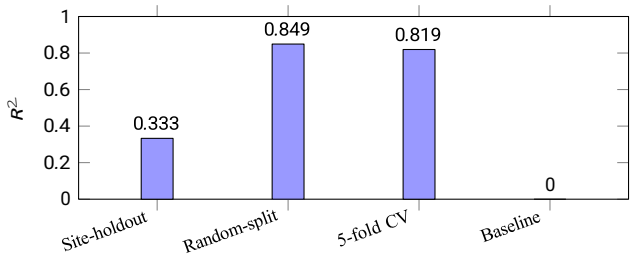


Fig. 4. R^2 across crowd-model evaluation protocols. The gap between site-holdout and random-split reflects the harder task of predicting crowd levels at entirely unseen destinations.

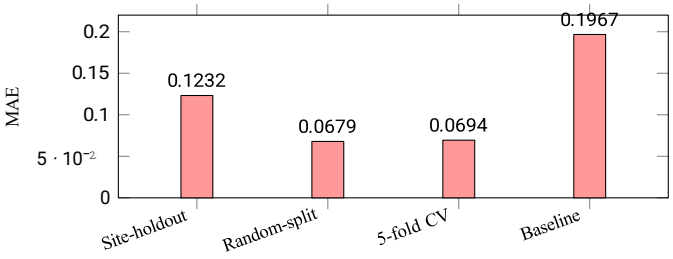


Fig. 5. MAE across crowd-model evaluation protocols. Lower is better; the model outperforms the mean-predictor baseline in every protocol.

TABLE IV
CROWD-PREDICTION MODEL PERFORMANCE (RIGOROUSLY EVALUATED)

Protocol	MAE	R^2	Acc.	Bal. Acc.
Site-holdout	0.1232	0.3331	77.31%	57.54%
Random-split	0.0679	0.8492	79.10%	80.00%
Baseline	0.1967	0.0000	—	—

Table V gives the per-class classification report for the crowd risk level on the random-split test set (n = 28,802), included to substantiate the balanced-accuracy gap noted in Section IV-D. Fig. 6 shows the same three metrics grouped by class.

TABLE V
RANDOM-SPLIT CLASSIFICATION REPORT (CROWD RISK LEVEL)

Class	Precision	Recall	F1	n
High	0.57	0.87	0.69	3,510
Medium	0.67	0.72	0.69	8,603
Low	0.95	0.81	0.88	16,689

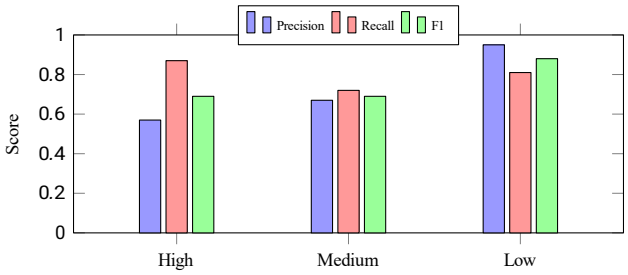


Fig. 6. Per-class precision, recall, and F1 for crowd risk level (random-split test set, n = 28,802).

Table VI summarises the testing approach and disclosed status per module.

TABLE VI
VALIDATION APPROACH AND EVALUATION STATUS ACROSS MODULES

Module	Reported Result	Status
SOS	92.4% usefulness, 8.2 s latency (simulated, n=200)	Simulation-stage
Itinerary	88.7% satisfaction, 74.3% adherence (simulated)	Simulation-stage
Wellness	Dosha RF 81.0% / F1 0.794 vs. SVM 52.4%, KNN 38.1%	Validated vs. baselines
Crowd / Risk	MAE 0.1232, R^2=0.3331, Acc. 77.31%	Validated (site-holdout)

VI. DISCUSSION AND LIMITATIONS

TourCeylon's contribution is the integration of four previously siloed concerns under a shared architecture. We deliberately distinguish two evaluation tiers in this paper: Module 1 and Module 2 metrics are reported from an architecture-level simulation (n = 200) intended to stress-test system wiring and integration points, not to substitute for measured production performance; Module 3 is supported by a genuine comparative model evaluation and ablation study; and Module 4 is evaluated under the field's more demanding site-holdout protocol, directly measuring generalisation to destinations absent from training. The gap between crowd-module site-holdout (R^2=0.3331) and random-split (R^2=0.8492) performance, visualised in Fig. 4, illustrates why random-split evaluation is an unreliable proxy for real-world crowd-prediction deployment, and we report both rather than the more favourable figure alone. The per-class results (Table V, Fig. 6) show recall for the High-risk class (0.87) exceeding precision (0.57), indicating the model is tuned toward over-flagging rather than under-flagging high-crowd conditions — a reasonable operating point for a safety-adjacent alerting use case, disclosed here rather than left unexamined. We consider the explicit separation of simulation-stage and rigorously-validated results, applied consistently rather than blended into uniform headline numbers, to be part of this paper's contribution.

VII. FUTURE WORK

Planned work includes replacing Module 1 and Module 2 simulation-stage KPIs with measured evaluations against real or field-piloted user feedback; conducting a formal data-protection impact assessment for the QR-based onboarding and SOS location-handling design (Section IV-E); expanding automated test coverage for the wellness module; and investigating site-independent meteorological and demand-proxy features as a likely contributor to the crowd module's site-holdout performance ceiling.

VIII. CONCLUSION

This paper presented TourCeylon, a four-module integrated tourism intelligence platform for Sri Lanka. Rather than reporting uniform metrics of equal evidentiary weight across all modules, we explicitly separate simulation-stage architecture KPIs (SOS, itinerary) from rigorously validated empirical results (wellness model comparison and ablation; crowd site-holdout evaluation), arguing that this distinction, consistently disclosed, is itself a necessary contribution for tourism-analytics systems built on comparably constrained data.

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